

The essence of calculus

Three big ideas, discovered from one circle.

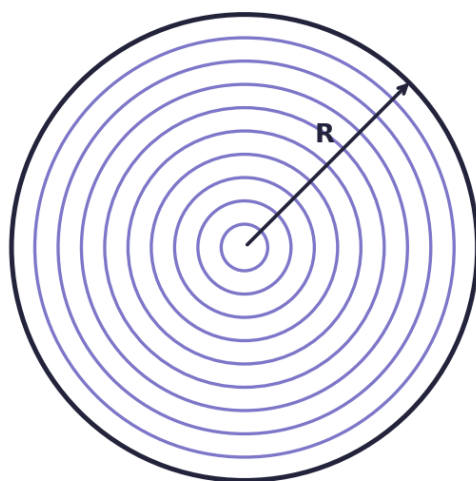
THE IDEA

Calculus is not a pile of formulas to memorise — it is three connected ideas, and you can stumble into all three by asking one simple question: **why is the area of a circle πR^2 ?**

THE CIRCLE ARGUMENT

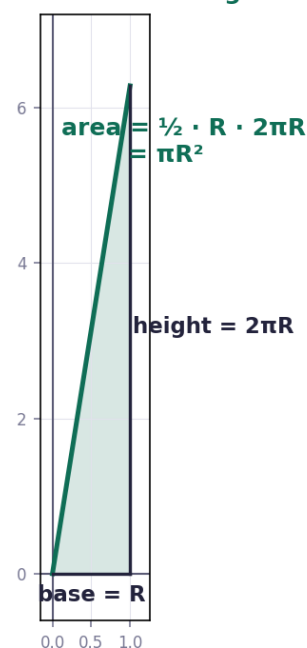
Slice the circle into very thin rings. One ring at radius r has thickness dr and length $2\pi r$, so its area is roughly $2\pi r \cdot dr$. Now unroll all the rings and stand them side by side: they form a triangle with base R and height $2\pi R$. Its area is $\frac{1}{2} \cdot R \cdot 2\pi R = \pi R^2$.

slice the circle into thin rings



each ring $\approx$ a thin rectangle

unroll them $\rightarrow$ a triangle



THE THREE IDEAS HIDING IN THAT ARGUMENT

Idea	What it is
Integral	Adding up infinitely many tiny pieces to get a total
Derivative	How fast a quantity is changing at an instant
They are opposites	Integrating and differentiating undo each other — the fundamental theorem

The strategy behind all of calculus: **approximate a hard problem with many easy pieces, then make the pieces infinitely small.**

WORKED EXAMPLE

CHAPTER 1 — WORKED EXAMPLE

the three big ideas, seen in one picture

THE CIRCLE

one ring at radius r : thickness dr , length $2\pi r$

$$\text{ring area} \approx 2\pi r \cdot dr$$

ADD ALL THE RINGS (this is an INTEGRAL)

stack them $\rightarrow$ a triangle with base R and height $2\pi R$

$$\text{area} = \frac{1}{2} \cdot R \cdot 2\pi R = \pi R^2$$

THE THREE IDEAS

Integral

adding up infinitely many tiny pieces

Derivative

how fast a quantity changes

They are opposites the fundamental theorem of calculus

Why it matters for ML: training a model means asking "if I nudge this weight slightly, how much does the loss change?" That is a derivative — and every weight in the network has one.

FORMULAS FROM THIS CHAPTER

Ring area	$2\pi r \cdot dr$
Circle area	πR^2
Integral	add up infinitely many tiny pieces
Derivative	rate of change at an instant

TEST — Chapter 1

1. A ring at radius $r = 3$ has thickness $dr = 0.1$. Estimate its area using $2\pi r \cdot dr$. $\rightarrow 2\pi(3)(0.1) \approx 1.88$
2. Using the triangle picture, what is the area of a circle with $R = 4$? $\rightarrow \frac{1}{2} \cdot 4 \cdot 2\pi(4) = 16\pi \approx 50.3$
3. In one word each: what does an integral do, and what does a derivative do? $\rightarrow$ **integral: accumulates** · **derivative: measures change**